CLOSURES OF CONNECTED SETS AND THE DISJOINT-UNION DESCRIPTION OF THE CONNECTED COMPONENTS

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- The definition of a connected topological space, of a connected component and of a connected subset. Set up in the author's section on connected components, immediately before the counted statement. Fixes the objects of the four numbered assertions.

Lemma 0.1. *Let X be a topological space.*

(1) *If $T \subset X$ is connected, then so is its closure.*
(2) *Any connected component of X is closed (but not necessarily open).*
(3) *Every connected subset of X is contained in a unique connected component of X.*
(4) *Every point of X is contained in a unique connected component, in other words, X is the disjoint union of its connected components.*

Proof. Let $\overline{T}$ be the closure of the connected subset T. Suppose $\overline{T} = T_1 \amalg T_2$ with $T_i \subset \overline{T}$ open and closed. Then $T = (T \cap T_1) \amalg (T \cap T_2)$. Hence T equals one of the two, say $T = T_1 \cap T$. Thus $\overline{T} \subset T_1$. This implies (1) and (2).

Let A be a nonempty set of connected subsets of X such that $\Omega = \bigcap_{T \in A} T$ is nonempty. We claim $E = \bigcup_{T \in A} T$ is connected. Namely, E is nonempty as it contains Ω. Say $E = E_1 \amalg E_2$ with E_i closed in E. We may assume E_1 meets Ω

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(after renumbering). Then each $T \in A$ meets E_1 and hence must be contained in E_1 as T is connected. Hence $E \subset E_1$ which proves the claim.

Let $W \subset X$ be a nonempty connected subset. If we apply the result of the previous paragraph to the set of all connected subsets of X containing W, then we see that E is a connected component of X. This implies existence and uniqueness in (3).

Let $x \in X$. Taking $W = \{x\}$ in the previous paragraph we see that x is contained in a unique connected component of X. Any two distinct connected components must be disjoint (by the result of the second paragraph).

To get an example where connected components are not open, just take an infinite product $\prod_{n \in \mathbf{N}} \{0, 1\}$ with the product topology. Its connected components are singletons, which are not open. $\square$